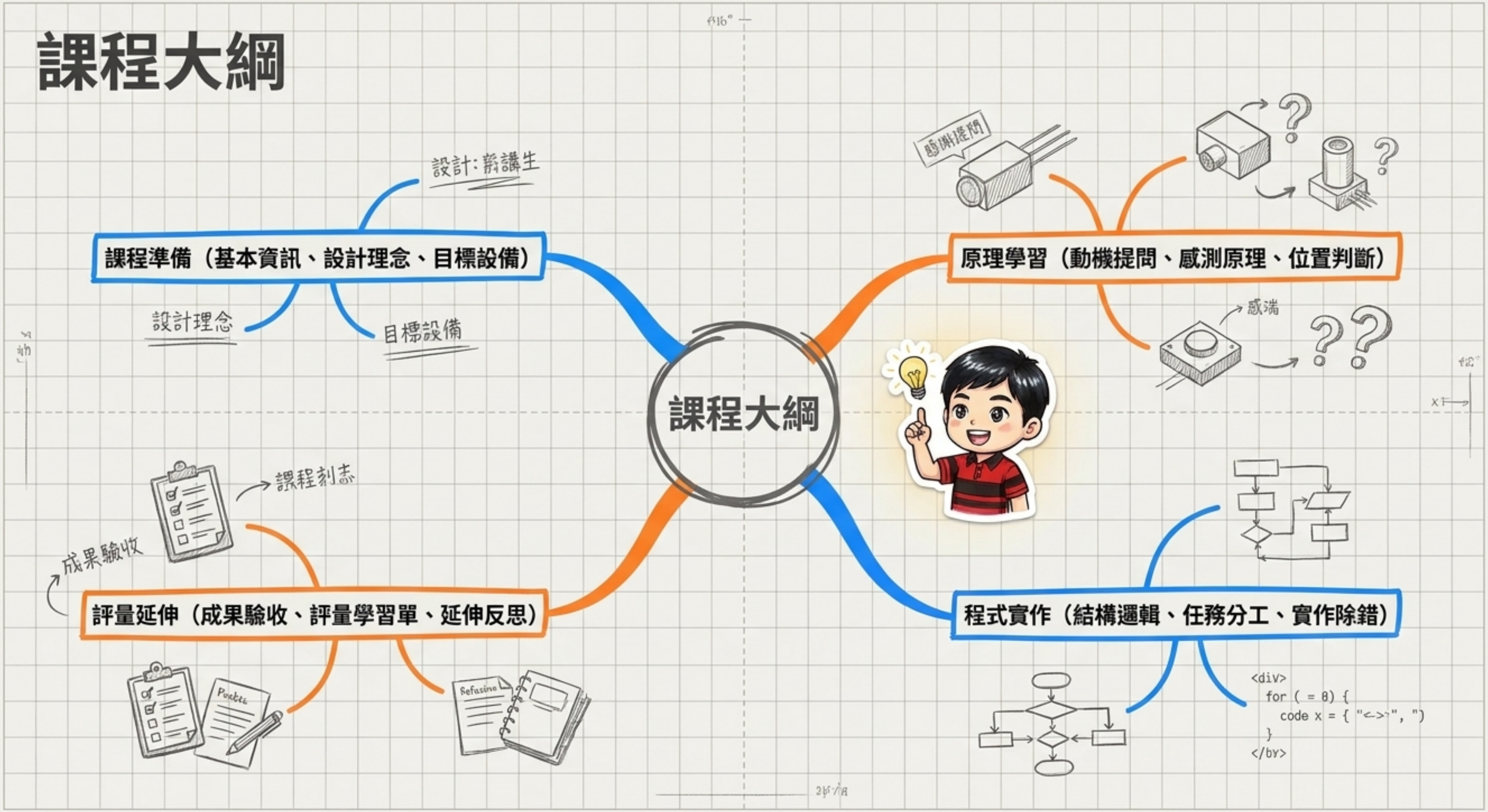
循跡達人：機器人程式實戰

帶領機器人自己學會走路

課程大綱


課程大綱
課程準備（基本資訊、設計理念、目標設備）
設計理念
目標設備
評量延伸（成果驗收、評量學習單、延伸反思）
成果驗收
課程剖析
原理學習（動機提問、感測原理、位置判斷）
感測
程式實作（結構邏輯、任務分工、實作除錯）

for (= 0) {
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}
</bv>

課程基本資訊

七年級資訊
科技領域

單節課四
四十五分鐘

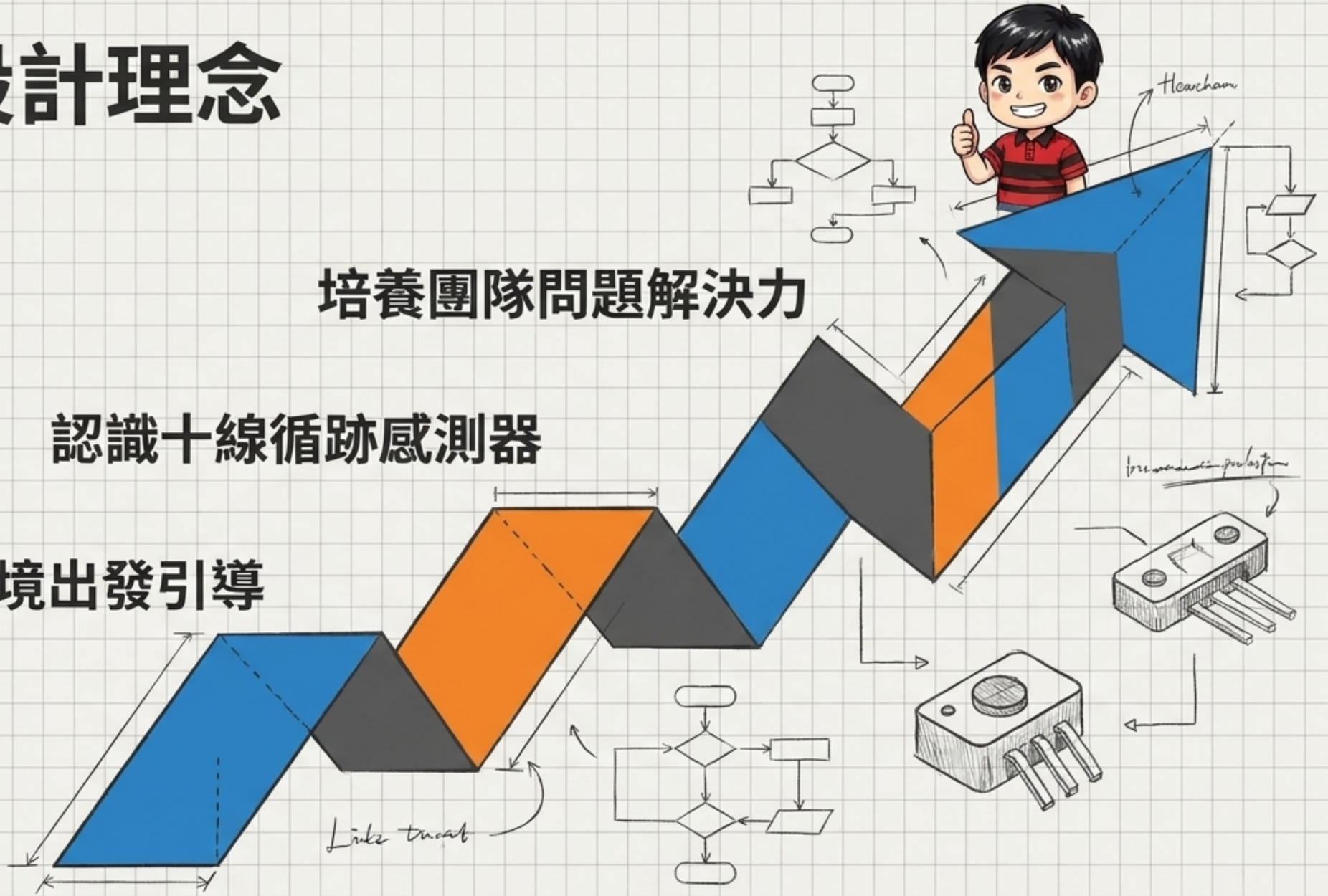
實作與合作
作學習為主

課程設計理念

培養團隊問題解決力

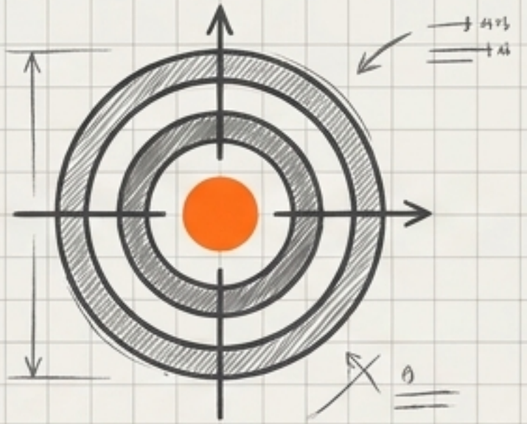
認識十線循跡感測器

從生活情境出發引導

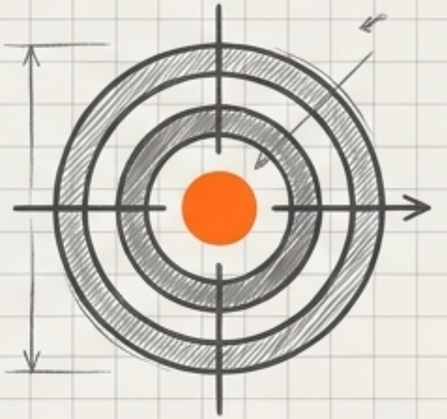


核心學習目標

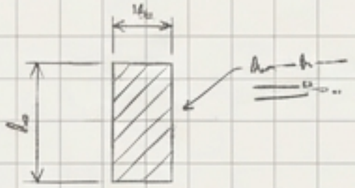
理解感測器運作原理



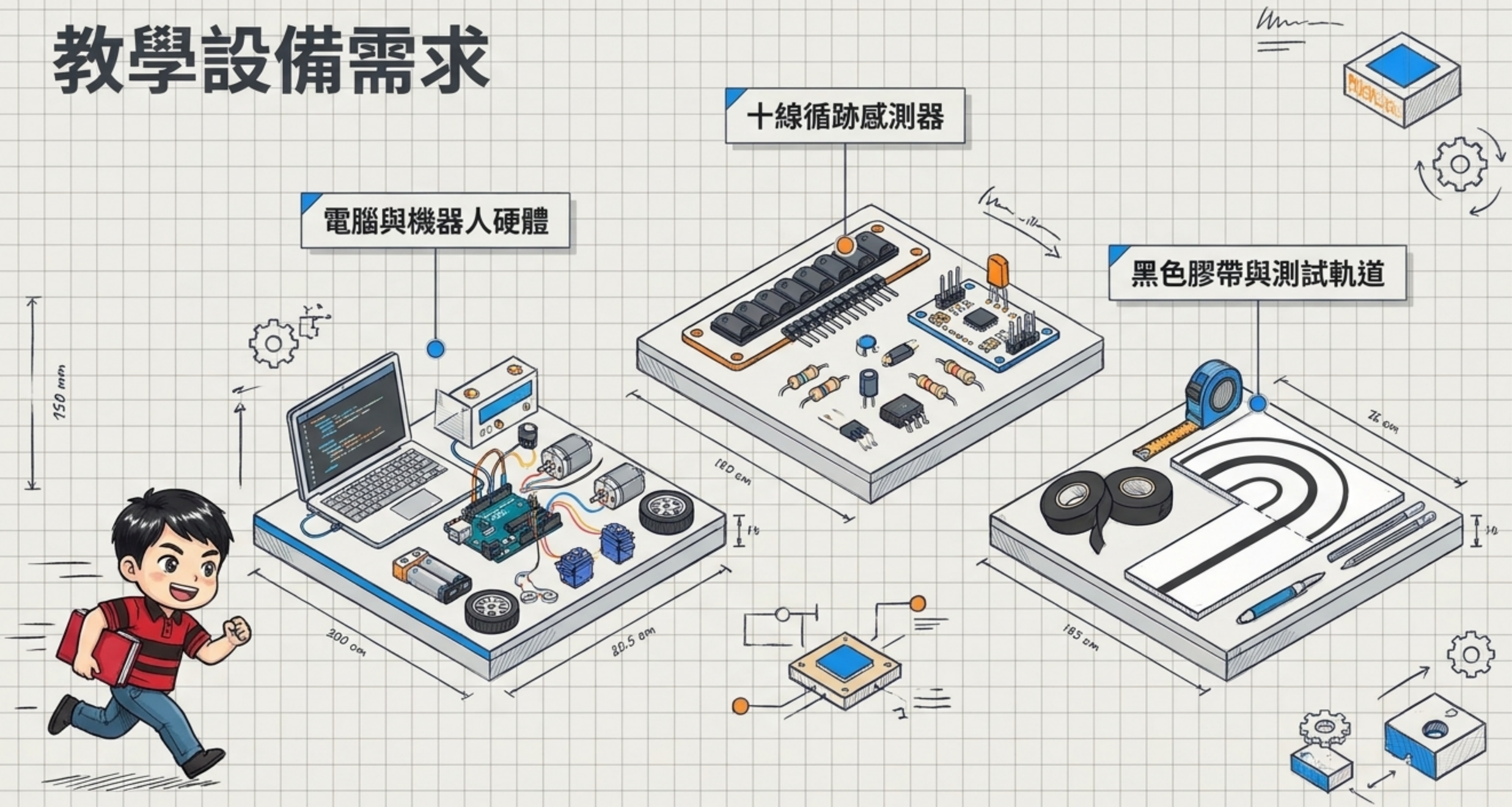
運用程式控制邏輯



團隊合作完成循跡



教學設備需求



課前軟硬體整備

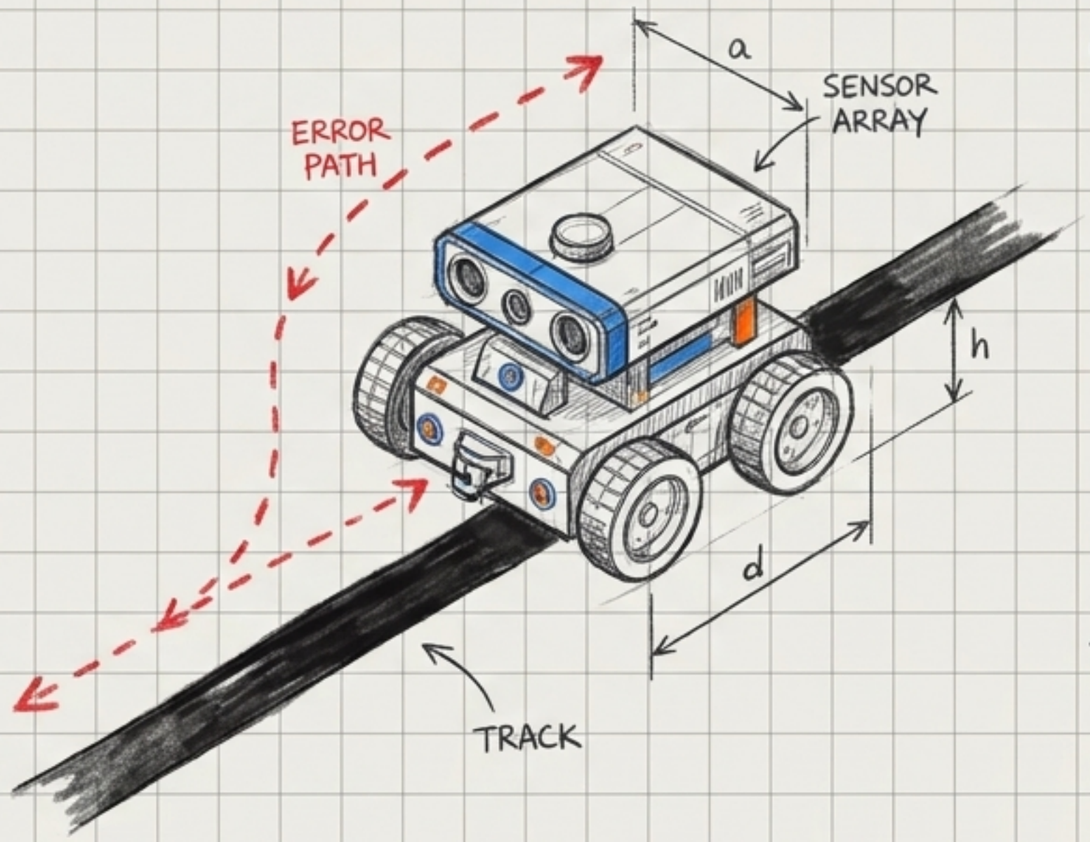
- [x] 測試硬體運作正常
- [x] 確認程式編輯環境
- [x] 學生具備基礎操作力

引起動機與提問

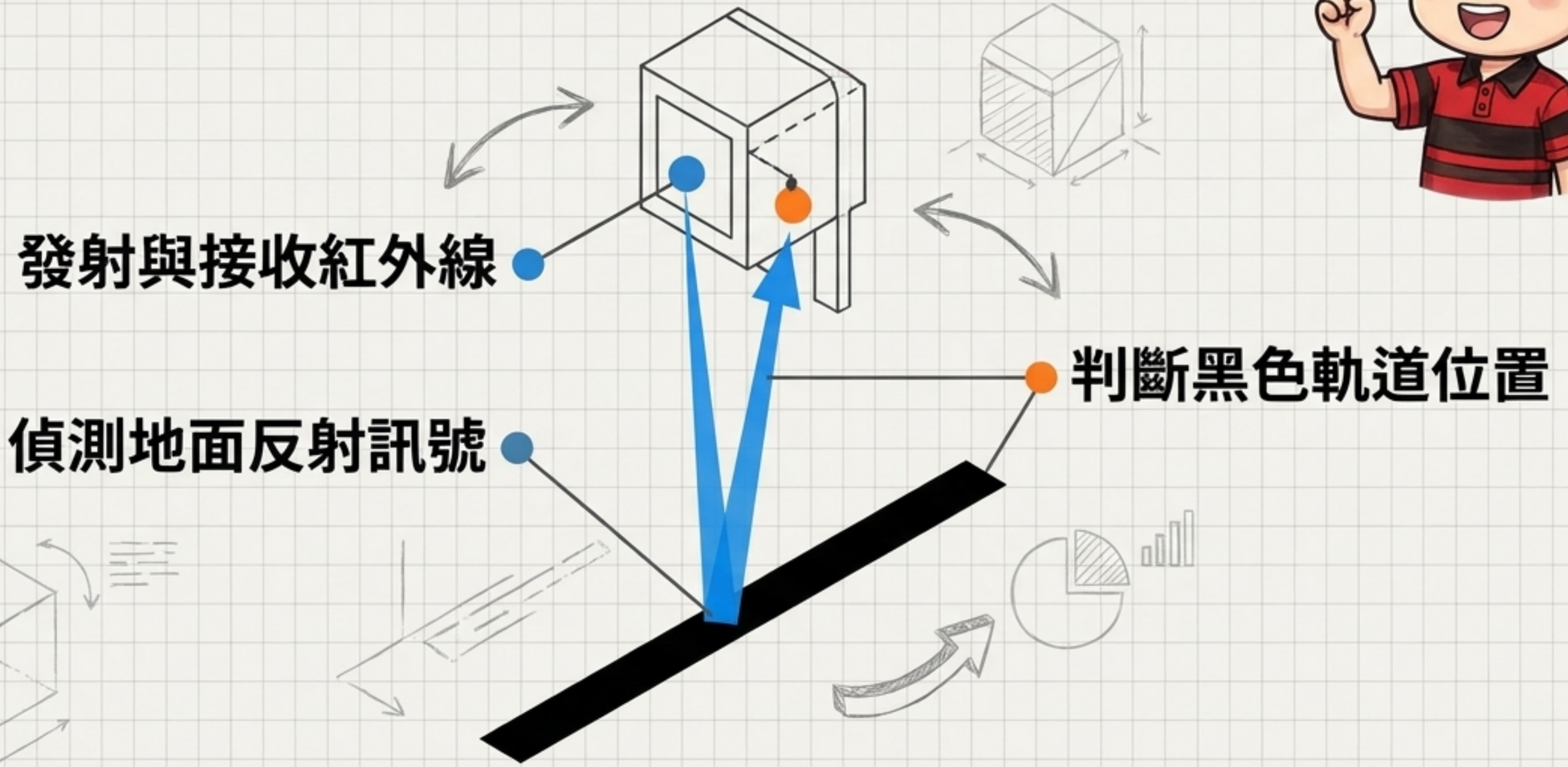
機器人怎能沿黑線走

走歪了如何知道方向

觀察感測與程式控制

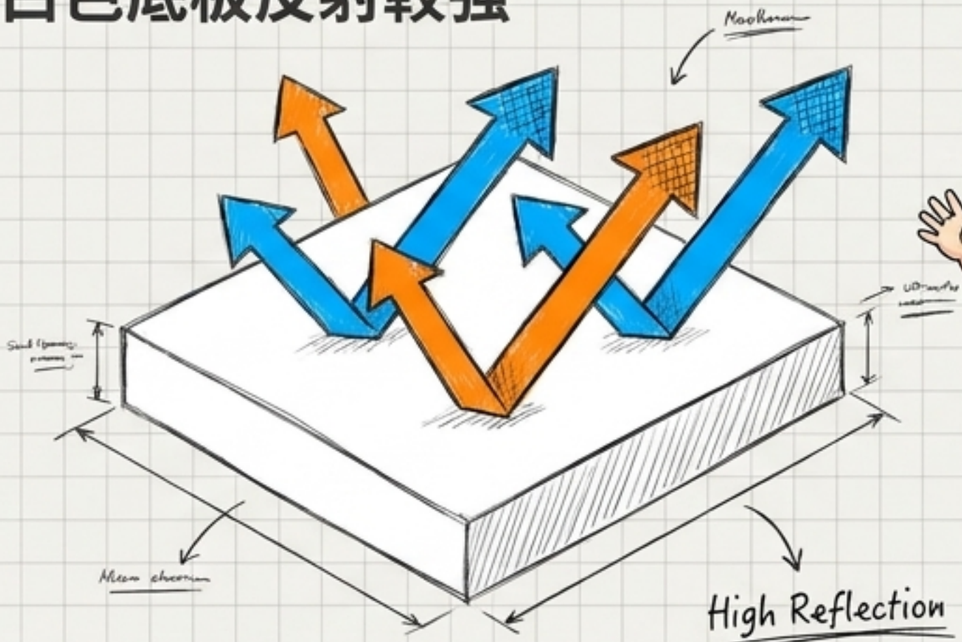


感測器運作原理

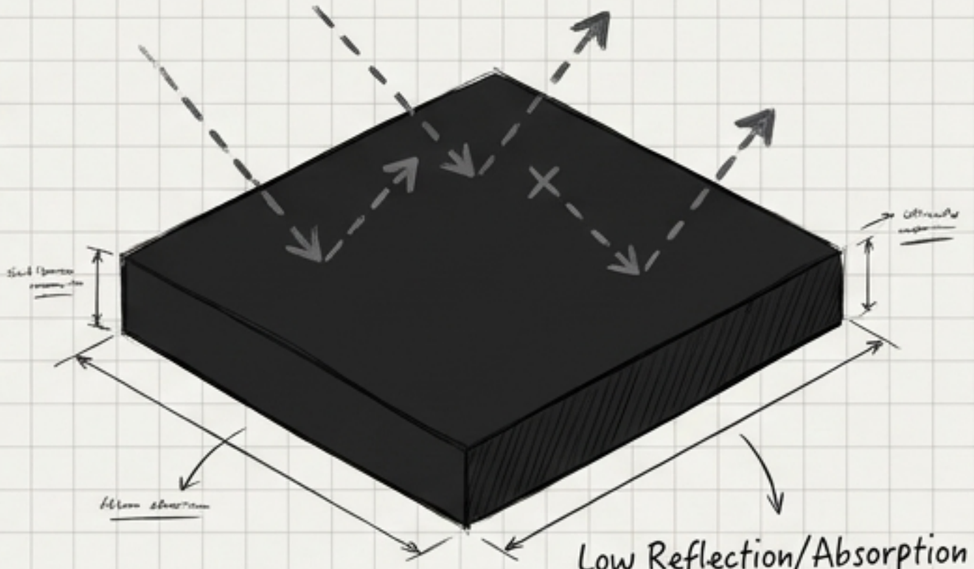


黑白反射差異

白色底板反射較強

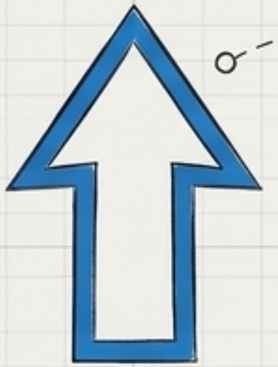


黑色膠帶吸收較多

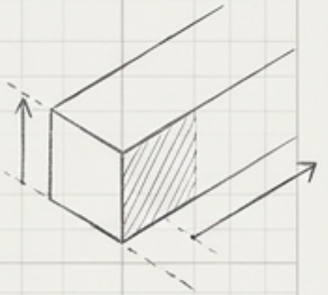


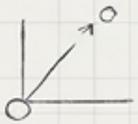
藉此區分背景與目標

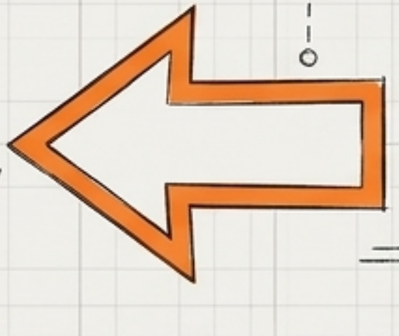
感測器位置判斷



黑線置中維持直行

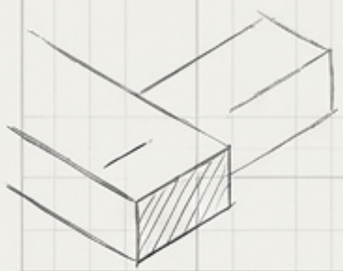




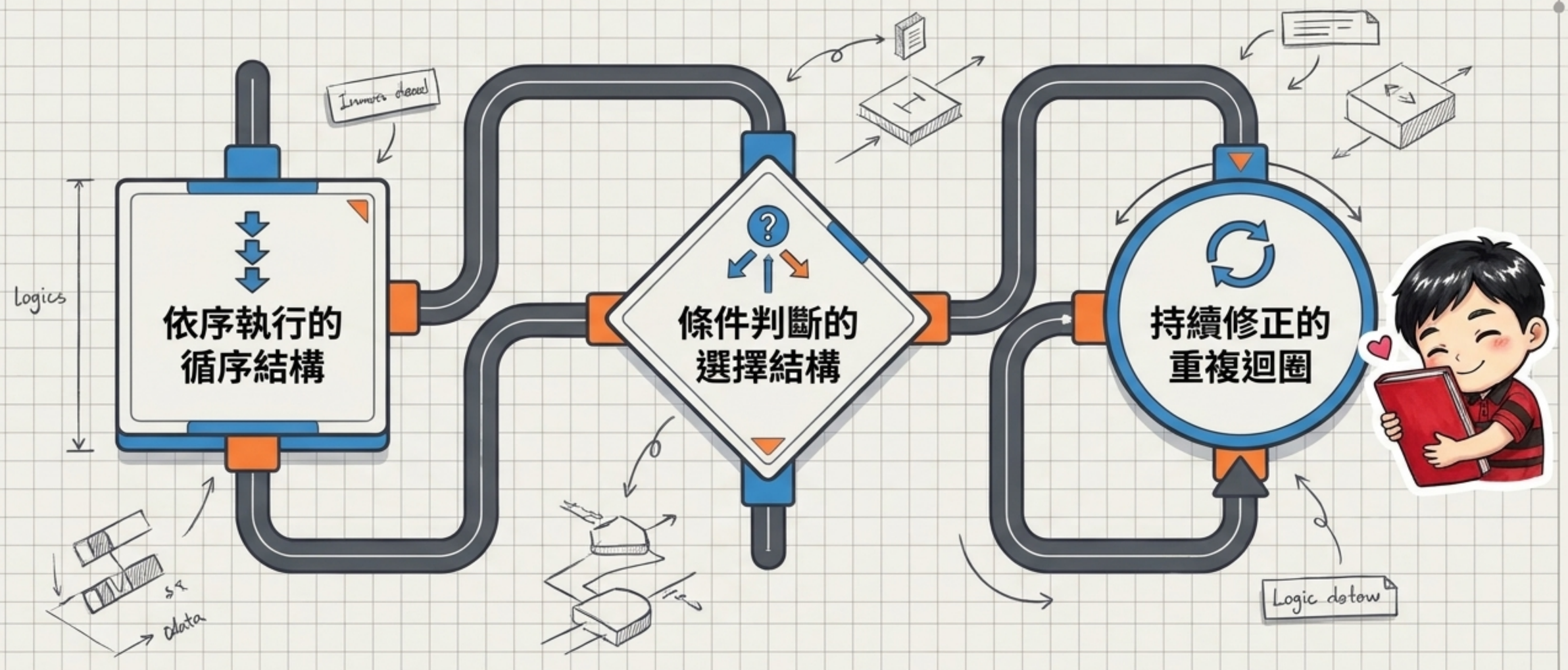


偏左則向左方修正

偏右則向右方修正



三大程式結構

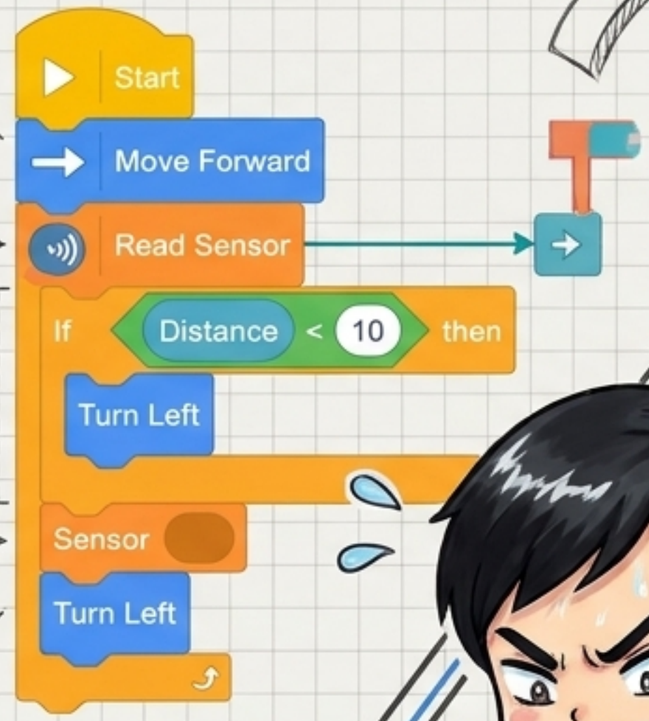


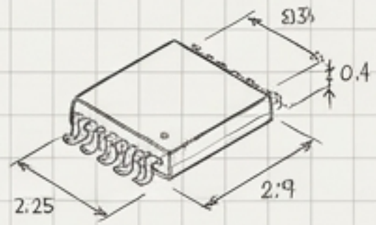
程式與硬體連接

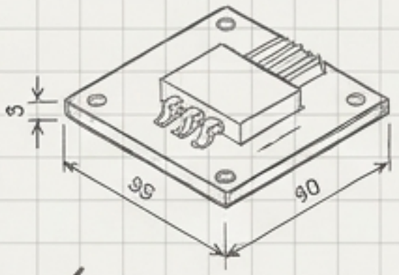
示範軟體介面操作

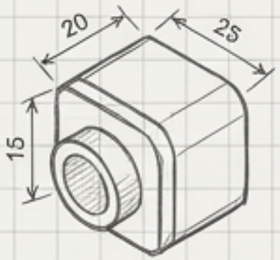
組合感測控制積木

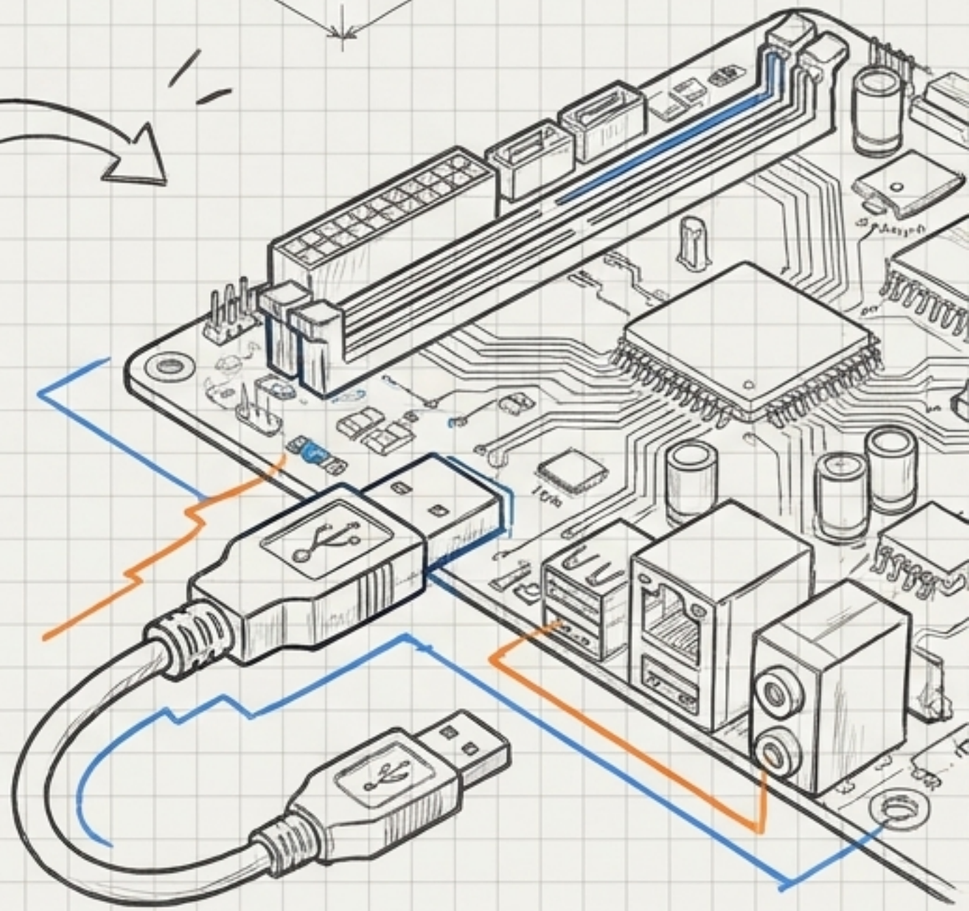
接線測試讀取數值











任務分工合作

實作與程式除錯

機器人放軌道測試

修改判斷參數除錯

觀察偏移調整速度

```
if (sensorValue > threshold) {
  adjustSpeed(-5);
}
```

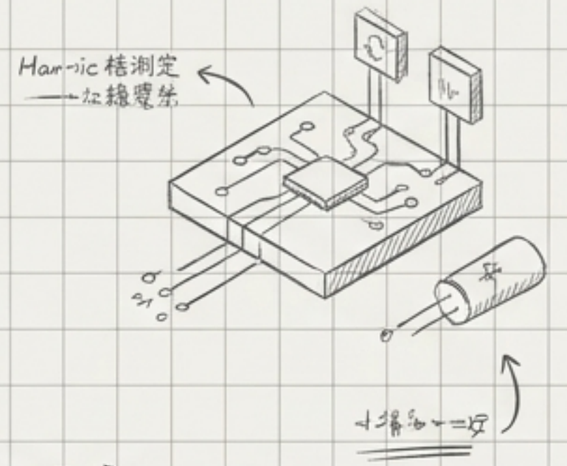
```
DEBUG:
Path deviation
detected.
...
```

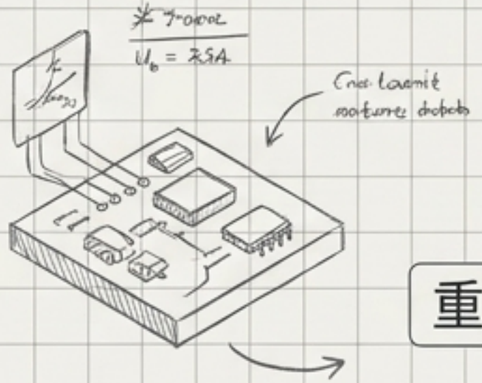
Tech Lab Sketchbook

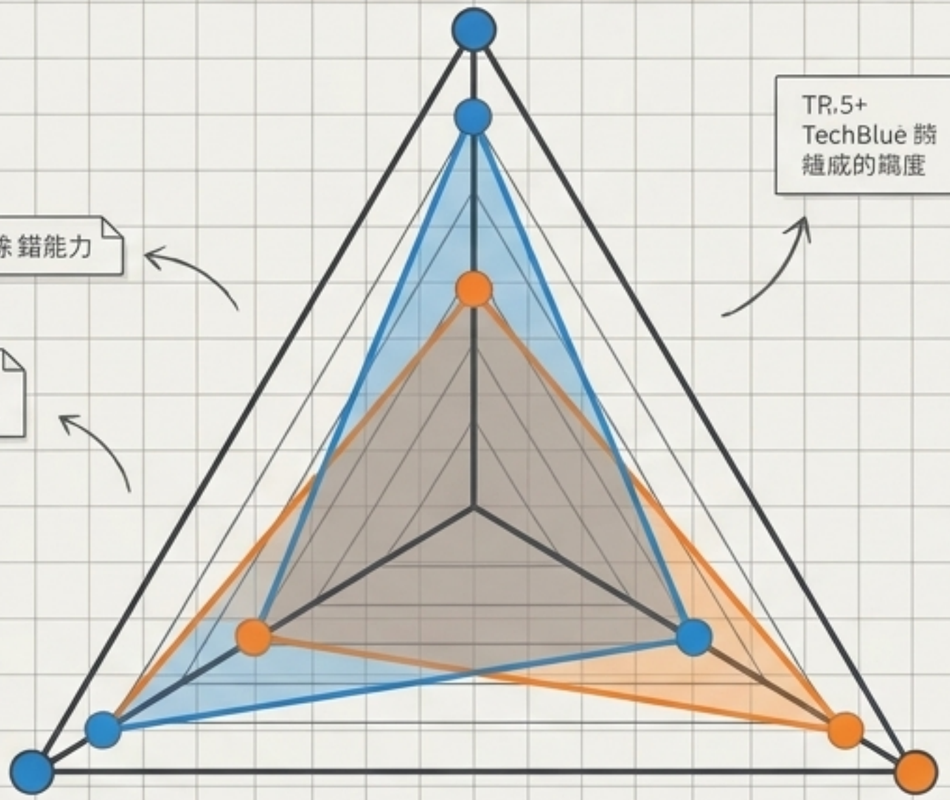
成果驗收與反思

實作評量規準

評量原理與程式邏輯

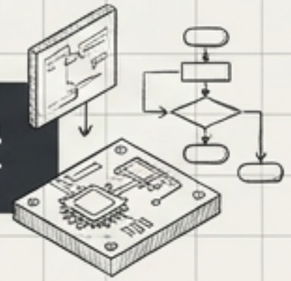






學習單重點紀錄

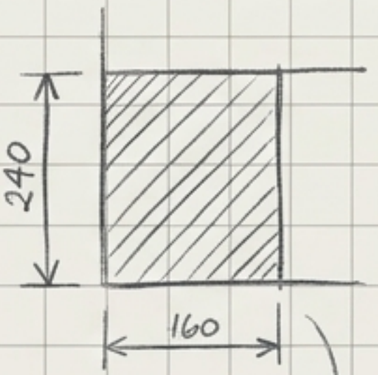
寫下原理與控制邏輯



勾選測試與除錯結果

紀錄困難與學習反思

課前檢查清單


240
160

ON
ON / Ready
確保設備
電量充足
ON
ON / Ready
軌道與軟
硬體就緒
ON
ON / Ready
教師預先
測試程式

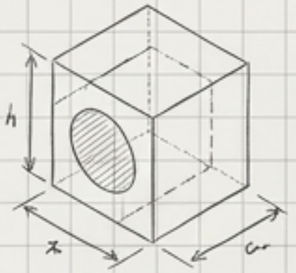
延伸學習挑戰

不同角度彎道循跡

比較速度與穩定度

斷線尋回策略設計

教師教學省思

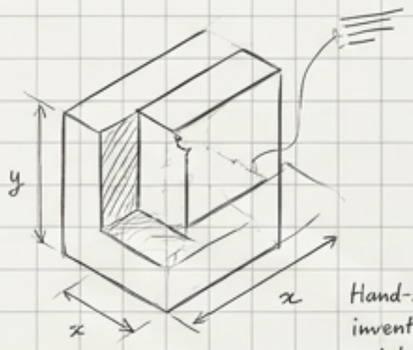


Tech Blue
data

Instructional annotation in mentorship

- 學生是否理解原理
- 紀錄實作遭遇困難
- 評估時間與改進處

An stractic annotcont



Hand-sitchén inventor's notebook

Data Node
Data